EDA Brainstorm: Opportunities for TruBridge

Trubridge Health Analytics Externship

Context

TruBridge provides Revenue Cycle Management (RCM), Electronic Health Record (EHR), coding, and Patient Engagement solutions and services to community and rural hospitals and clinics across the U.S. Its core value proposition is removing administrative and financial complexity for smaller, often underserved healthcare providers. The ideas below brainstorm where Exploratory Data Analysis could surface hidden patterns in TruBridge's own data or its clients' data, the same way EDA uncovered Spotify's listening-habit patterns or the retailer's unexpected B2B buying trend.

Brainstormed EDA Opportunities

1. Claims Denial Patterns (Revenue Cycle Management)

Question: Are certain claim types, payers, or hospital departments producing denials at a much higher rate than others?

Why it matters: Like the retailer's surprise B2B buying spike, a spike in denials for a specific code or payer might reveal a systemic coding or documentation issue rather than random noise -- something a monthly denial-rate report alone might miss.

2. Time-to-Payment Variation Across Rural vs. Larger Clients

Question: Do smaller, rural hospital clients experience longer billing-to-payment cycles than larger clients, and if so, at which specific step does the delay occur?

Why it matters: TruBridge's stated focus is rural/community providers -- understanding where cash flow bottlenecks concentrate could directly inform their predictive cash-flow dashboards.

3. EHR Usage Patterns by Time of Day / Staff Role

Question: Are there specific times of day, days of the week, or staff roles where EHR system usage drops, errors spike, or support tickets cluster?

Why it matters: Mirrors the Spotify example directly -- just as skip rates varied by time of day, EHR friction may follow a time-based or role-based pattern that a simple usage-count metric would hide.

4. Coding Accuracy Trends After AI/ML Tool Rollout

Question: Since introducing AI-assisted coding, has accuracy actually improved uniformly, or only for certain code categories or client sizes?

Why it matters: A top-line 'accuracy improved X%' number could mask uneven adoption or uneven benefit across different hospital types.

5. Patient Engagement Feature Adoption vs. Patient Outcomes

Question: Do patients who actively use TruBridge's patient engagement/portal tools show different follow-up or readmission patterns than those who don't?

Why it matters: This connects directly to the externship's broader theme -- social determinants of health and access may explain who does or doesn't engage with these tools in the first place, not just tool design.

6. Geographic Clustering of Client Financial Health

Question: Do TruBridge's rural hospital clients in certain states or regions consistently show weaker financial metrics (e.g., slower collections, higher denial rates), independent of hospital size?

Why it matters: Ties back to the exact kind of county-level, geography-driven analysis explored in this externship's own asthma/air-quality project -- regional context often explains patterns that aggregate numbers hide.

7. Support Ticket Text Analysis

Question: Are there recurring themes or phrases in client support tickets that point to a shared root cause, rather than many unrelated one-off issues?

Why it matters: A straightforward ticket-count metric answers 'how many,' but EDA (e.g., simple text/word-frequency analysis) could answer 'why' -- similar to how Spotify needed to go past raw skip-counts to find the real behavioral pattern.

Reflection

This task recommended pairing with a peer to widen the brainstorm, but I worked through it independently. I focused on ideas that map onto TruBridge's three real business segments (RCM, EHR, Patient Engagement) rather than generic EDA examples, and included one idea (#6) that deliberately echoes the geographic/public-health analysis approach used earlier in this externship, since that framework could transfer directly to a healthcare analytics company's own client base.